

DEVELOPING AN INNOVATION ECOSYSTEM OF KAZAKHSTANI UNIVERSITIES AS A TOOL FOR COMMERCIALIZING INTELLECTUAL PROPERTY

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ABSTRACT

As Kazakhstan transitions to a knowledge- and innovation-based economy, the development of effective mechanisms for commercializing intellectual property (IP) at universities is of particular importance. The development of innovation ecosystems is a key tool for ensuring the transfer of technologies, knowledge, and competencies from academia to the real economy. This article examines the development of innovation ecosystems at Kazakhstani universities, analyzes institutional and organizational barriers limiting the commercialization of intellectual property, and suggests ways to overcome them. A comparative analysis with practices of leading universities in the European Union and the United States is conducted, revealing successful models for integrating academic science, business, and government within the framework of the Triple Helix concept. Particular attention is paid to the role of technology transfer centers, startup incubators, and endowment funds in ensuring sustainable innovative development at Kazakhstan's universities. The results obtained can be used to formulate university strategies and public policy in science, innovation, and higher education.

Keywords: university innovation ecosystem; commercialization; intellectual property; technology transfer; entrepreneurial university; innovation infrastructure.

INTRODUCTION

The current development of the global economy is characterized by the increasing role of knowledge, innovation, and digital technologies as key factors in sustainable growth. Under these conditions, universities are no longer solely educational institutions and are becoming active participants in the innovation ecosystem, ensuring the transfer of scientific developments to the real sector of the economy. For Kazakhstan, striving to build a diversified, innovation-driven economy, the development of an effective system for the commercialization of RIAs at universities is becoming a strategic priority of public policy.

Despite the availability of scientific potential and government support, Kazakhstan's universities still face a number of systemic challenges that hinder the effective commercialization of innovations. These include weak links between academic science and business, underdeveloped innovation infrastructure, limited financial and managerial instruments for technology transfer, and the absence of sustainable mechanisms to motivate researchers and faculty to engage in entrepreneurial activity.

At the same time, the international experience of universities in the US, UK, Germany, the Netherlands, and Finland demonstrates that successful commercialization of intellectual property is only possible within a well-established innovation ecosystem, where the university, business, and government effectively interact within the framework of the "triple helix" model. These ecosystems include technology transfer centers, business incubators, venture funds, research consortia, and entrepreneurship departments, ensuring a closed innovation cycle—from idea to marketable product.

Thus, the relevance of this study stems from the need to adapt best international practices for developing university innovation ecosystems to the conditions of Kazakhstan. The aim of this study is to analyze the current state and development prospects of university innovation ecosystems in Kazakhstan as a tool for commercializing the results of intellectual activity. To achieve this goal, the following objectives were set: to determine the institutional and organizational characteristics of Kazakhstani university ecosystems; to identify barriers hindering the commercialization of innovations; and to substantiate the development directions of university ecosystems within the context of national innovation policy.

Literature Review. In modern scientific literature, an innovation ecosystem is viewed as a set of institutions, interacting entities, and infrastructure elements aimed at the creation, development, and commercialization of scientific knowledge. Within this system, the university plays a key role as an intermediary between scientific research, industry, and government. According to the triple helix model, effective innovation is achieved through close integration of academia, business, and government [1].

The open innovation paradigm emphasizes the need for the exchange of ideas and technologies between the university and the external environment. Research shows that the sustainable development of university ecosystems depends on their ability to ensure the influx of knowledge and resources from external sources while simultaneously disseminating research results throughout the economy and society [2]. In this regard, the development of university technology transfer centers, incubators, and startup platforms is becoming an important element of the innovation development system [3].

In the European Union, universities are viewed as strategic participants in the innovation ecosystem, capable of contributing to the sustainable development of regions.

The practice of European universities shows that the effectiveness of interactions between universities, businesses, and government directly depends on the quality of research infrastructure, specialist competencies, and regulatory support. Research by the European University Association emphasizes that European universities are becoming hubs of sustainable innovation ecosystems that foster technological and social progress [3]. Similar trends are observed at US universities, where innovation is integrated into educational and research processes, and the commercialization of scientific results is supported through venture capital financing and an entrepreneurial culture [4,5].

In Kazakhstan, the development of university innovation ecosystems is in its infancy. Research shows that the potential of domestic universities in technology transfer remains underutilized. An analysis by Albekova et al. identified institutional barriers and a lack of managerial competencies that hinder the effective commercialization of scientific developments [6]. Commercialization centers and technology transfer offices have been established at several universities in Kazakhstan, including Nazarbayev University, Al-Farabi Kazakh National University, Satbayev University, and AlmaU, providing administrative, legal, and consulting support to researchers [7, 8, 9]. These centers aim to find investors, protect intellectual property, and promote university developments to the market. However, experts believe their potential has not yet been fully realized due to limited financial resources and insufficient private sector involvement [10].

Key barriers remain inadequate links between academic science and industry, a weak intermediary infrastructure, regulatory restrictions, and low motivation for researchers to engage in entrepreneurial activity. International experience demonstrates that the formation of a sustainable innovation ecosystem requires a combination of institutional incentives, developed human capital, and government support. For Kazakhstan, adapting best international practices and integrating them into the national education and science system can become the basis for increasing the effectiveness of commercializing the results of university intellectual activity.

ETHODS

The methodological framework for this study is based on a combination of systems, institutional, and comparative analytical approaches. The systems approach allowed us to examine the university's innovation ecosystem as a holistic structure, encompassing interconnected elements such as research departments, technology transfer centers, entrepreneurial initiatives, venture funds, and government support institutions. The institutional approach was used to analyze the legal framework governing the commercialization of intellectual property and to identify mechanisms for university interaction with businesses and the government within Kazakhstan's national innovation system.

To achieve this goal, we employed content analysis, comparative analysis, and expert assessment. Content analysis was used to examine the development strategies of leading universities in Kazakhstan and the regulatory documents governing their innovation activities. Comparative analysis allowed us to compare Kazakhstani practices with the experience of universities in the European Union and the United States, where innovation ecosystems have already reached a high level of institutional maturity. An expert assessment method was used to analyze the opinions of science management specialists, representatives of commercialization offices, and the academic community.

The study's information base included official documents from the Ministry of Science and Higher Education of the Republic of Kazakhstan, reports from the National Agency for Innovation Development, university activity data posted on their official websites, and materials from international organizations such as the OECD, the European University Association, and the World Intellectual Property Organization. In addition, the results of studies published in peer-reviewed journals such as Scopus and Web of Science for the period 2019–2024, devoted to the commercialization of scientific research and the development of university ecosystems, were used.

The results of the analysis allowed us to identify key structural and institutional features of the innovation ecosystem of Kazakhstan's universities and to determine the factors that facilitate or hinder the effective commercialization of intellectual property.

RESULTS AND DISCUSSION

The development of innovation ecosystems at Kazakhstan's universities reflects the overall vector of transformation of the national higher education system, focused on the integration of science, education, and business. Over the past decade, the country has been creating legal and organizational conditions that encourage universities to commercialize the results of their intellectual property. The Ministry of Science and Higher Education operates grant support mechanisms and the "Commercialization of Science" program, aimed at increasing the practical impact of scientific research. However, analysis shows that the level of development of university ecosystems remains uneven and depends on the size of the university, available resources, and the involvement of regional partners [6], [8].

The most developed ecosystem elements are found in large national universities. For example, Nazarbayev University and Al-Farabi Kazakh National University have technology transfer centers, business incubators, and intellectual property management structures. Satbayev University and Almaty Management University are implementing startup incubators and acceleration programs for students and young researchers. At the same time, regional universities have limited access to funding and human resources, which hinders their innovative activity (see table 1).

Table 1 - Comparative analysis of the state of university innovation ecosystems

Indicator	Kazakhstan	European Union	USA
Availability of technology transfer centers	Established at leading universities (10–15%)	Established at almost all universities	A mandatory element of research universities
Innovation Financing	Predominantly public, limited	Mixed: public and private	Predominantly private, venture capital
Number of university-based startups	Less than 5% of research projects	20–25% of projects	30–40% of projects
Share of patents commercialized through licenses	3–5%	25–35%	More than 40%
Venture capital funds at universities	Isolated cases	Common in innovation clusters	Standard practice
Entrepreneurial Culture in Universities	Emerging	Institutionalized	Deeply integrated into the educational process
Note: Compiled by the authors based on the analysis of open sources and data [11,12,13].			

The obtained results show that Kazakhstani universities are in a transitional stage in the formation of innovation ecosystems. Despite the existence of a legal framework and individual successful examples, the systemic nature and institutional maturity of these ecosystems are still insufficient. The main factors behind this lag are limited mechanisms for engaging private capital and the weak development of university entrepreneurship.

At universities in the European Union, a key element is the integration of research, education, and entrepreneurial functions through strategic partnerships with businesses. European universities are developing multi-level support for innovation: from laboratories and incubators to regional innovation clusters interacting with local governments. A distinctive feature of the American model is the high share of venture capital financing, the active participation of private investors and endowment funds, as well as a culture of risk and innovative freedom for researchers [3], [5].

For Kazakhstan, the identified differences indicate the need to strengthen three areas of development. First, institutional consolidation is required: the formation of a unified strategy for university ecosystems at the national level, taking into account regional specificities. Secondly, it is important to strengthen the system of interaction between universities and industry, ensuring mutual benefit through contract research, joint laboratories, and technology platforms. Thirdly, it is necessary to develop a sustainable entrepreneurial culture within the academic environment, motivating researchers to commercialize scientific ideas and developing competencies in innovation management.

Analysis shows that commercialization activity in Kazakhstan is gradually increasing. According to reports

from the Ministry of Science and Higher Education, the number of projects supported through commercialization programs more than doubled from 2020 to 2024, yet their total share in national R&D remains below 10 percent. This indicates significant growth potential remains. The most dynamically developing areas are digital technologies, renewable energy, and biotechnology, where university research finds practical application in industry and agriculture.

A comparison with international practices confirms that the critical factor in the effectiveness of innovation ecosystems is the availability of sustainable financial instruments and a flexible organizational management structure. At universities in the EU and the US, technology transfer offices enjoy autonomy in decision-making, enabling them to quickly respond to business and investor demands. In Kazakhstan, similar structures currently have limited powers and insufficient resources. This necessitates a revision of intellectual property management mechanisms and a strengthening of the role of universities as economic entities capable of generating income from innovation.

Human capital is particularly important. Sustainable ecosystems require specialists with expertise in project management, technology marketing, patent law, and venture capital financing. Kazakhstani universities are gradually introducing educational programs in innovation management and entrepreneurship, but their share of total personnel training remains low. Developing human resource potential is key to achieving the strategic objectives of science commercialization.

The analysis concluded that the innovation ecosystem of Kazakhstani universities is in its infancy and is showing positive development trends. Its further improvement

requires systemic measures to strengthen the technology transfer infrastructure, stimulate the entrepreneurial activity of researchers, and expand business participation in university projects. In the long term, such measures will enhance the competitiveness of the national economy and integrate Kazakhstan into the global knowledge and innovation space.

Conclusion. The results of the study confirmed that the formation of an innovative ecosystem of Kazakhstani universities is a prerequisite for the effective commercialization of intellectual property and enhancing the competitiveness of the national economy. Current trends in the development of higher education and science demonstrate the transition of universities from their traditional role as knowledge centers to active participants in economic processes, generating innovative solutions and creating added value.

Kazakhstani practice demonstrates some successful examples of building university-type ecosystems; however, their institutional maturity and sustainability remain limited. The main problems remain the insufficient integration of science and business, the fragmentation of the innovation infrastructure, the weak participation of the private sector in research funding, and the low motivation of researchers to commercialize their own developments.

A comparative analysis with practices in the European Union and the United States allowed us to identify key factors that ensure the successful development of university innovation ecosystems. These include the presence of autonomous technology transfer offices, a developed system of venture and endowment financing, government support for entrepreneurial initiatives, and an institutional culture of innovative thinking. Adapting these mechanisms to Kazakhstan's national conditions could accelerate the commercialization of scientific developments and increase the attractiveness of universities as centers of innovative growth.

Priority areas for improving university ecosystems include the institutionalization of technology transfer centers, the development of public-private partnership mechanisms, the implementation of training programs for specialists in innovation management and entrepreneurship, and the creation of incentives for researchers and faculty to engage in innovative projects.

In the long term, the development of Kazakhstan's university innovation ecosystem should lead to the formation of a sustainable national model for knowledge commercialization integrated into the global science and technology space. This will create the foundation for strengthening the country's technological sovereignty and increasing the effectiveness of interactions between universities, businesses, and the government.

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